Title: Do geofence warrants violate the Fourth Amendment?

Early Geofence History

Geofencing technology was invented in the early 1990s and later patented in 1995 by American inventor Michael Dimino. It relied on brand-new Global Positioning System (GPS) and Global System for Mobile Communication (GSM) technology for a tracking system to locate objects anywhere on the globe. Originally it was designed for things like fleet management and tracking vehicles and equipment, not criminal investigations.

Geofence Technology

You are most likely using a service that requires location accessibility; think Google Maps or social media. If so, your phone constantly broadcasts its location through a combination of GPS satellites, nearby WiFi networks, and cell towers, even when you're not actively using it. Google receives these phone signals and logs them with a precise timestamp and coordinates, building a continuous record of where your device has been. According to NBC News, approximately 74% of smartphone users fall into this category. The "geofence" itself is just a virtual boundary, essentially a shape drawn on a map defined by coordinates, such as a circle or square. When a device crosses the drawn boundary, Google's servers detect it because they're constantly comparing your device's reported location against whatever boundaries have been set up. It all happens behind the scenes in Google's servers, massive databases that store billions of location pings from hundreds of millions of devices, all indexed by time and coordinates so they can be searched and filtered instantly.

Geofence & Investigations

A geofence warrant works like a digital dragnet in three steps. First, police draw a specific virtual shape around a crime scene and ask Google for the data of every person whose phone was in that area during a certain time window. Then, Google hands over anonymous ID numbers, in which police examine and flag anyone who seems suspicious based on their movements. Finally, police ask Google to unmask the names and account information of the people they flagged. The big concern is step one. Police don't know who they're looking for yet. They're essentially demanding data on everyone who walked by, including totally innocent people.

Geofence Warrants Produced

Google reported that it had received 982 geofence warrants in 2018, 8,396 in 2019, and 11,554 in 2020. As our world becomes more reliant on technology, one can only assume this number has increased over the past couple of years. A 2021 transparency report showed that 25% of all data requests from law enforcement to Google were geofence data requests. Google has pushed back claiming they don't hold the type of information police are looking for. However, an easy alternative for police is relying on other applications, such as Apple, Lyft, Snapchat, Uber, or Microsoft.

The Ambiguity Dilemma

The fact that law abiding citizens can and do get swept up in geofence warrants raises a strong dilemma. Consider step two mentioned above—police seek out suspicious patterns such as someone who entered and left the area at the exact time of the crime or who arrived, stopped briefly, then left quickly. Additionally, police scan for unusual behavior, including a device that turned off right at the scene and turned back on nearby shortly after, or someone who took an odd or evasive route away from the scene, all of which could simply be movements that reflect nothing more than an individual's daily habits. The inherent problem with this process is that behavior flagged as suspicious may have entirely innocent explanations—a person retracing their route to retrieve a forgotten item, or a device that lost power coincidentally near the crime scene. While such occurrences may seem unlikely, the stakes of the legal system demand a higher standard. When real lives and liberties are under threat, no degree of reasonable doubt can be dismissed.

The Legal Debate

The complex debate here lies within the Fourth Amendment's ban on unreasonable government searches. In this case, and numerous similar instances, the question is whether or not the government has a right to "search first" in hopes of spotting potential suspects of a crime.

Real World Example / Chatrie v. U.S.

The Supreme Court is now hearing the case Chatrie v. United States, which could decide the future of geofence use in police investigations. Chatrie was convicted of bank robbery near Richmond, Virginia in 2019. While a federal judge agreed with the defense's statement regarding the broad language of the warrant, the evidence was admitted due to believing the officer was acting in good faith—an exception to the exclusionary rule. This means that an individual does something with honest intentions and a sincere belief that what they're doing is right or lawful; even if it later turns out to be wrong. With the geofence evidence brought into court, Chatrie was found guilty and appealed the decision. The U.S. Court of Appeals for the Fourth Circuit also found Chatrie guilty since he voluntarily turned on Google location services; finding the evidence entirely lawful under the Fourth Amendment. Remember, the Fourth Amendment says the government cannot search you, your home, or your belongings without a good reason and, usually, a warrant. To receive a warrant the police or investigators must show probable cause. In Chatrie's case, defense lawyer Adam G. argues the geofence warrant allows the government to "search first and develop suspicions later."

Latest News on Chatrie v. U.S.

On April 27th, 2026, the Supreme Court heard opening statements and oral defense arguments for Chatrie v. United States. Opening statements are the first thing each side says at the start of a trial or hearing, before any evidence is shown or arguments are made. Oral arguments are a live back and forth conversation between lawyers and judges about the legal issues in a case.

Probable Outcomes

The court seems to be heading in one of three directions: dismiss or affirm narrowly on good faith, decide on particularity grounds only, or rule a broader Fourth Amendment ruling. A broad Fourth Amendment ruling is unlikely, due to the court being required to directly address whether location history has Fourth Amendment protection at all; something all justices want to avoid. Dismiss or affirm narrowly on good faith would essentially lead to the same outcomes seen in both the district court and appeals court. Ruling on particularity grounds appears to be most favorable. Simply put, instead of a warrant asserting "search whatever you want and take whatever looks suspicious," it would declare "search apartment 4B at 123 Main Street for a stolen red bicycle." Notice the strong detail, which is not what most geofence warrants currently possess. A particularity ground ruling would restrict Google—or other location service platforms—from being the sole candidate of narrowing searches, putting that power back into the hands of a judge. This would also set precedent—past court decisions that future courts must follow—requiring warrants to spell out criteria at each investigative step, without creating sweeping new Fourth Amendment doctrine.